

PAEC (Philippine Aeronautical English Corpus)

Overview

PAEC is a corpus-based training and analysis platform designed to help pilots and air traffic controllers improve their aviation English communication skills in compliance with ICAO standards. Built around authentic ATC communications from RPLL (Ninoy Aquino International Airport), the platform delivers interactive training modules, automated dialogue analysis, and progress tracking through a modern web interface using Next.js 16 and PostgreSQL.

Features

Training Modules

- 4 Comprehensive Training Exercises
 - Scenario-Based Simulation: Full ATC clearance exchanges across APP/DEP, GND, and RAMP phases
 - Readback/Hearback Correction: Identify non-standard or incorrect pilot phraseology
 - Jumbled Clearance: Drag-and-drop word ordering into correct ICAO sequence
 - Pronunciation Drill: Practice ICAO standard pronunciation for numbers, letters, and aviation terms
- Concept-Level Answer Matching with intelligent equivalence scoring
- Immediate ICAO Doc 4444 Feedback on every submission
- Phase-Aware Detection across Approach/Departure, Ground/Taxi, and Ramp/Pushback

Analysis Engine

- Automated Transcript Analysis for PDF, DOCX, and plain text uploads
- ICAO Compliance Reporting with violation categorization
- Non-Standard Phraseology Detection aligned with the thesis taxonomy
- Number Pronunciation Error Identification
- Safety-Critical Pattern Recognition
- PDF and CSV Export of analysis results
- Phase-Aware Engine for APP/DEP, GND, and RAMP communications

Dashboard & Progress Tracking

- Session History across all training modules
- Performance Score Tracking with trend analysis
- Per-Category Progress Breakdowns
- Visual Charts showing improvement over time
- Per-Question Feedback Records within each session

Audio & Speech

- Web Speech API speech recognition for pronunciation drills
- Text-to-Speech playback of ATC clearances and readbacks
- Browser-Native Audio with no third-party dependencies
- Real-Time Pronunciation Feedback against ICAO standards

User Management

- Email and Password Registration with email verification
- Google OAuth Sign-In and account linking
- Forgot Password and Reset Password flows
- Secure JWT Sessions via NextAuth.js 5
- First-Login Onboarding flow
- Profile Management and account settings
- Account Deletion with full data removal

UI/UX Design

Design Principles

- Modern Web Standards: Built with Next.js 16 App Router and React 19
- Accessibility First: Semantic HTML, keyboard navigation, and proper contrast
- Responsive Layout: Adapts to desktop, tablet, and mobile screen sizes
- Smooth Animations: Framer Motion for polished transitions and interactions

Theme System

- Tailwind CSS: Utility-first styling with consistent design tokens
- Aviation-Inspired Branding: Color scheme aligned with aeronautics
- Custom Typography: Optimized for readability of long ATC transcripts
- Consistent Spacing: Systematic layout grid across all pages

Security & Privacy

Data Protection

- Bcrypt Password Hashing: Industry-standard salted password storage
- JWT Session Tokens: Secure, stateless authentication via NextAuth.js
- Email Verification: Confirms account ownership before granting access
- Authentication Required: All training and analysis endpoints require valid auth
- Input Validation: Client and server-side validation on all API routes

Privacy Features

- Minimal Data Collection: Only essential account and session information
- Per-User Data Isolation: Sessions and progress scoped to the authenticated user
- Account Deletion: Users can permanently remove their account and all data
- Secure Communication: HTTPS/TLS for all network requests

Performance Optimizations

Efficiency Features

- Server Components: Reduced client-side JavaScript with React 19
- App Router Streaming: Progressive page rendering for faster perceived load
- Database Connection Pooling: Efficient PostgreSQL connections via pg
- Code Splitting: On-demand bundle loading per route
- Caching Strategy: Static and dynamic content caching at the edge

Resource Management

- Image Optimization: Next.js Image component with automatic format selection

- Document Parsing: Stream-based PDF and DOCX text extraction
- Serverless Database: Neon PostgreSQL for elastic scaling and cold-start efficiency
- Lazy Loading: Charts and heavy components loaded on demand

Browser Support

Minimum Requirements

- Browser: Chrome 90+, Firefox 88+, Safari 14+, or Edge 90+
- JavaScript: Enabled
- Network: Stable internet connection required
- Audio: Microphone access for the Pronunciation Drill module

Optimal Experience

- Browser: Latest Chrome or Edge for full Web Speech API support
- Display: 1280×720 or larger for the dashboard and analysis charts
- Audio: Speakers or headphones for text-to-speech playback
- Microphone: External or built-in microphone for pronunciation feedback

Acknowledgments

Technologies Used

- Next.js 16 — React framework with App Router
- React 19 — Modern UI library
- TypeScript 5.3 — Type-safe JavaScript
- Tailwind CSS — Utility-first styling
- PostgreSQL (Neon) — Serverless database
- NextAuth.js 5 — Authentication and session management
- Framer Motion — Animations and transitions
- @dnd-kit — Accessible drag-and-drop
- Recharts — Data visualization
- PDF.js — In-browser PDF parsing

- Web Speech API — Speech recognition and synthesis
- Nodemailer — Email delivery via Gmail SMTP

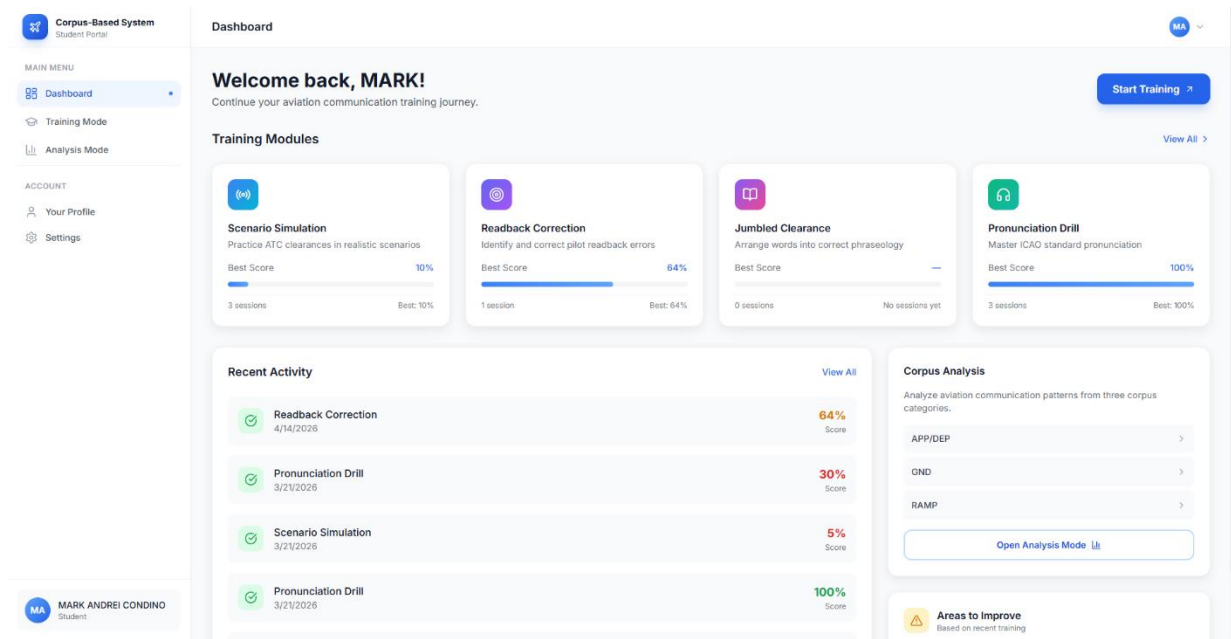
Corpus & Research

- PAEC Corpus v3.4 — Transcribed and annotated ATC recordings from RPLL
- Batch Lima — 31 student pairs from the BSAVCOMM 2nd-year program at the Philippine State College of Aeronautics (PhilSCA)
- Academic Year — 2nd Semester, AY 2024–2025
- Supervisor — Dr. Ramsey S. Ferrer
- Source — LiveATC.net recordings spanning February 1 to March 3, 2025
- ICAO Doc 4444 — Reference for standard aeronautical phraseology

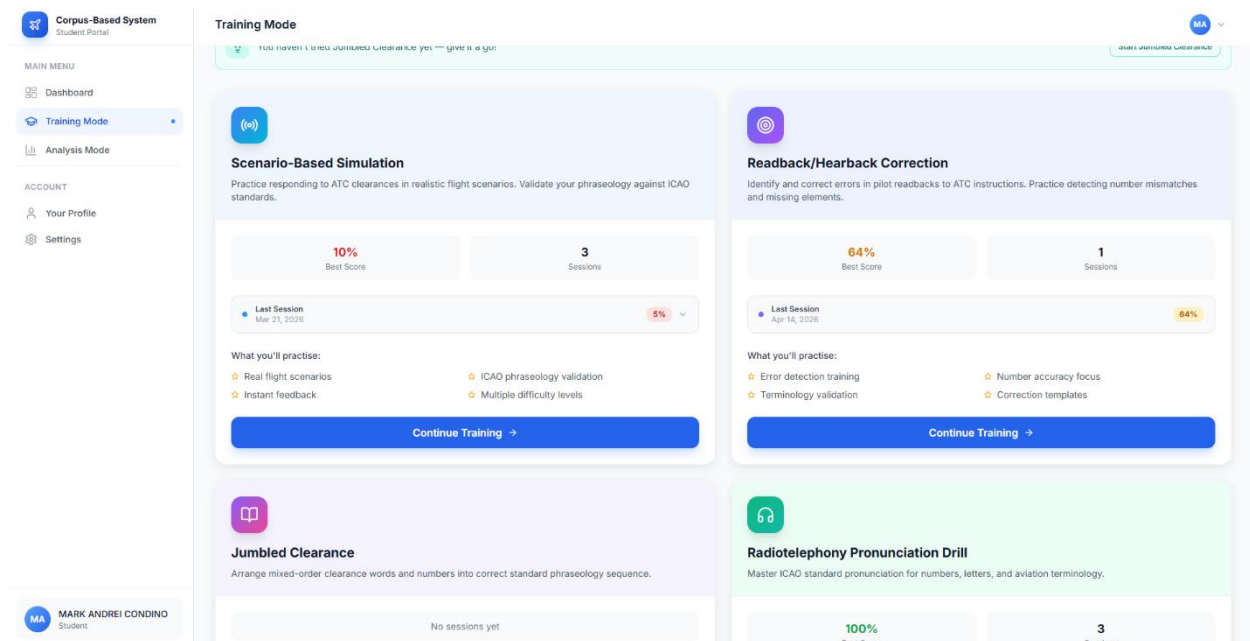
Page Overview

The screenshot displays the web application interface for "Master ICAO-Standard Phraseology". At the top, a navigation bar includes a logo, the title "Corpus-Based System", and links for "Features", "Training", "Analysis", and "About". A "Go to Dashboard" button is positioned on the right. Below the navigation bar, a section titled "Philippine Aeronautical English Corpus" introduces the main heading "Master ICAO-Standard Phraseology". A descriptive paragraph states: "Train with authentic pilot–ATC communications drawn directly from the Philippine Aeronautical English Corpus (PAEC). Improve your ICAO-standard phraseology and enhance aviation communication safety in Philippine operations." Two buttons, "Go to Dashboard" and "Explore Features", are provided. A "Sample Analysis" panel on the right shows an ATC instruction: "PAL456, descend and maintain four thousand, QNH one zero one three." and a corresponding Pilot Readback: "Roger, descend four thousand, QNH one zero one three, PAL456." Below the readback, two bullet points highlight non-standard elements: "Roger" (non-standard acknowledgement) and "maintain" (omitted from readback). A "Go to Dashboard" button is at the bottom of the panel. At the bottom of the page, it states "Built on ICAO Standards" and includes two status indicators: "ICAO Compliant" and "Research-Based".

HOME



DASHBOARD



Corpus-Based System
Student Portal

MAIN MENU

- Dashboard
- Training Mode
- Analysis Mode

ACCOUNT

- Your Profile
- Settings

MA MARK ANDREI CONDINO
Student

Training Mode

Training

Scenario-Based Simulation

Scenario 1 of 10

Philippine 415

Airbus A320

FLIGHT PHASE

Approach

CAVOK

On Approach

Situation: On short final for runway 24 at NAIA, Tower calls a go-around due to a slow aircraft on the runway.

ATC Clearance

Philippine Four One Five, go around, fly runway heading, climb Two Thousand.

Your Response

Type your readback response here...

Next Question

ICAO PHRASEOLOGY TIP
Always end your readback with your call sign. This confirms to ATC that the correct aircraft received and understood the instruction.

BackNext

TRAINING

Corpus-Based System
Student Portal

MAIN MENU

- Dashboard
- Training Mode
- Analysis Mode

ACCOUNT

- Your Profile
- Settings

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Analysis Mode

Analysis Mode

Analyze aviation communication patterns from the Philippine Aeronautical English Corpus.

Select Corpus Category

Approach/Departure Control

Communications between pilots and approach/departure control during climb-out and descent phases.

1,250 dialogues

45,000 words

Selected

Ground Control

Ground movement communications including taxi instructions and runway crossings.

980 dialogues

32,000 words

Ramp Control

Ramp and apron communications for aircraft parking and push-back operations.

720 dialogues

24,000 words

APP_DEP 2.pdf ✓ + Add Files

Analysis Results

APP/DEP Corpus

New AnalysisExport

29%

ICAO Score

24,386

Words

576

Exchanges

0.9

Non-std / %

Readback Analysis

51%

622

318

304

Error Breakdown

17

LANGUAGE

438

NUMBER

Corpus-Based System
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Analysis Mode

PROCEDURE

STRUCTURE

1031 total errors

Annotated Transcript

635 issues found across 2 conversations

PLT "Philippine", > **three** Zero Zero, Direct Cabatuan Climb Zero **five** Miles.

NUMBER

Number 'three' should be pronounced 'tree'

ICAO Doc 9432 Draft

YOU SAID

Three

SAY REPEAT

tree

CORRECTED TRANSMISSION

PHILIPPINE tree Zero Zero, Direct Cabatuan Climb Zero Five Miles.

ICAO requires specific number pronunciations to prevent misunderstandings over radio. "tree" is designed to be distinct and unambiguous, even with poor radio reception or background noise.

Why it matters:

Incorrect number pronunciation can cause altitude busts, heading deviations, or frequency errors - all potentially dangerous situations.

NUMBER

Number 'five' should be pronounced 'fife'

ICAO Doc 9432 Draft

1495 lines / 2 conversations

835 lines with errors

Exchange Analysis

574 exchanges analyzed

83%

574

TOTAL

188

DEPARTURE

233

APPROACH

0

HIGH-WEIGHT

cruise (153)

descent (143)

departure climb (137)

arrival (49)

initial departure (47)

approach (36)

final approach (1)

lineup (4)

1

Initial departure

Correct and complete

100%

2

Initial departure

100%

Corpus-Based System
Student Portal

MAIN MENU

- Dashboard
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- Analysis Mode

ACCOUNT

- Your Profile
- Settings

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Analysis Mode

Correct and complete

18

departure climb

Correct and complete

19

cruise

20

approach

Correct and complete

Show 20 more exchanges (554 remaining)

Number-Related Errors

438 total

Altitude mismatch

0

Heading errors

0

Speed discrepancy

0

Pronunciation notes

438

Squawk errors

0

Language-Based Errors

631 total

Incorrect Readback

0

Incorrect Hearback

1

Incomplete Readback

20

Incomplete Hearback

610

Key Findings

307 high-weight phraseology issues detected

Readback completeness: 51% (318/622)

Non-standard phraseology rate: 0.7 per 1,000 words

438 number-related errors detected

ISSUES DETECTED

- Running heading is not standard. Runway heading means maintain the magnetic heading of the runway. Confusion can cause serious misunderstandings.
- Decent (adjective) is not an aviation instruction. The correct term is descend.
- Stop instruction

Recommendations

Focus on complete readbacks: include altitude, heading, runway, and callign

Address most common issue: 'Okay' is not ICAO standard

Review ICAO number pronunciation (NINER, TREE, FIVE, FOW-ER)

Review APP/DEP procedures: climb/descent phraseology, approach clearances, vectoring

ANALYSIS